Bias and variance

* Bias : → Refers errors due to overly simplistic assumption made by the model.
  → Model with high bias tends to underfit the data.
  → perform poorly in both training and testing

* variance : → Refers errors due to model being more complex and sensitive to specific training data it has seen.
  → Model with high variance tends to overfit the data.
  → performs very well on training but poorly in test data.

Bias - variance tradeoff: Helps to balance complexity and the simplicity of the model.

i) High bias, low variance → underfitting

ii) low bias, high variance → overfitting

Trade off visualisation:

Total error = Bias^2 + Variance + Irreducible error

Boosting

Bagging – Reduces variance by training multiple models independently. (Reduces overfitting)

Boosting – Reduces both bias & variance by training the model sequentially. Each model tries to correct the errors made by previous model. This creats strong model from multiple weak models.

How it works ?

- Models are trained sequentially with each new model giving more importance to the data points that previous models misclassified.
- The predictions from all the models are combined, usually by a weighted sum / vote, where more weight is given to the models that performs better.

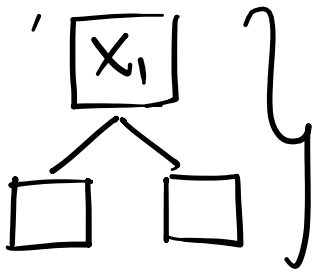
Examples:

1. AdaBoost (Adaptive Boosting)
2. Gradient Boosting
3. Extreme Gradient Boosting (XGBoost)

Adaboost: Ensemble learning algorithm designed to improve the performance of weak learners by combining them into stronger overall model.

⊕ Focus on misclassified datapoints on iterative training and adjust the mistake made by previous model.

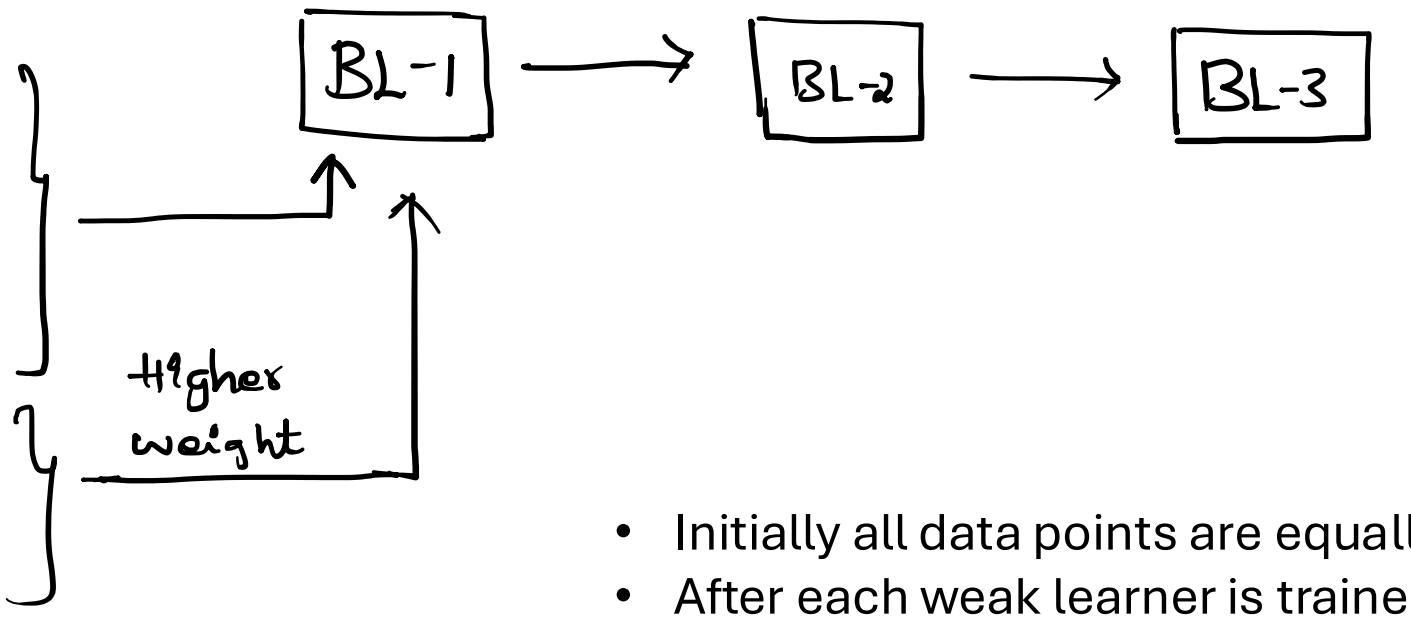
Weak learners: models that perform slightly better than random guessing. In adaboost decision stumps are the weak learners.



→ DT with a single split (max_depth = 1)

Sequential Training:

X1	X2	X3	Y
1			✓
2			✓
3			✓
.			✓
.			✗
.			✗
1000			✗



- Initially all data points are equally weighted.
- After each weak learner is trained Ada boost increase the weight of misclassified samples and decrease the weights of correctly classified.
- This forces the next weak learner to pay more attention to the hard to classify points.

How Adaboost works:

i) Initialization – Assign equal weights to all dp's.
If we have N samples, weight of each sample = $1/N$

ii) Training the weak learners.

iii) update the weights of records.

iv) Repeat (ii) & (iii) till predefined iteration.

v) Final model with aggregates of votes for classification or weighted outputs for regression.

⊛ Error Rate of Weak learner:

$$\rightarrow E_t = \sum_{i=1}^{n} w_i \times I\left(y_i \neq h_t(x_i)\right)$$

Where,

w_i - weights of samples i.

y_i - true label.

$h_t(x_i)$ - prediction from weak learner.

$I(y_i \neq h_t(x_i))$ - Indicator function that equal to 1 if the prediction is correct and 0 otherwise.

⊛ Learner Weight (α_t): Performance

⊛ Contribution of each weak learner to final prediction

$$\alpha_t = \frac{1}{2} \ln\left(\frac{(1-\epsilon_t)}{\epsilon_t}\right)$$

⊛ Weight update:

$$\Rightarrow w_i^{(t+1)} = w_i^{(t)} \cdot e^{\alpha_t \times I(y_i \neq h_t(x_i))}$$

⊛ If the prediction is correct in previous iteration weight will reduce in next iteration & vice versa.

X1	X2	X3	Y
1			
2			
3			
4			
5			
6			×
7			

Sample Weight

1/7

.

.

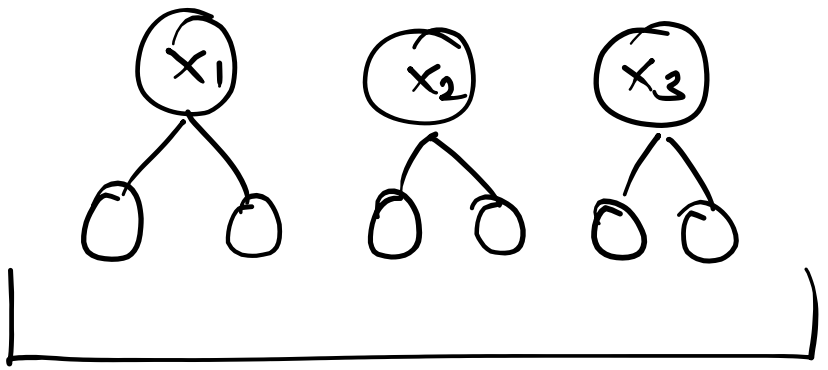
.

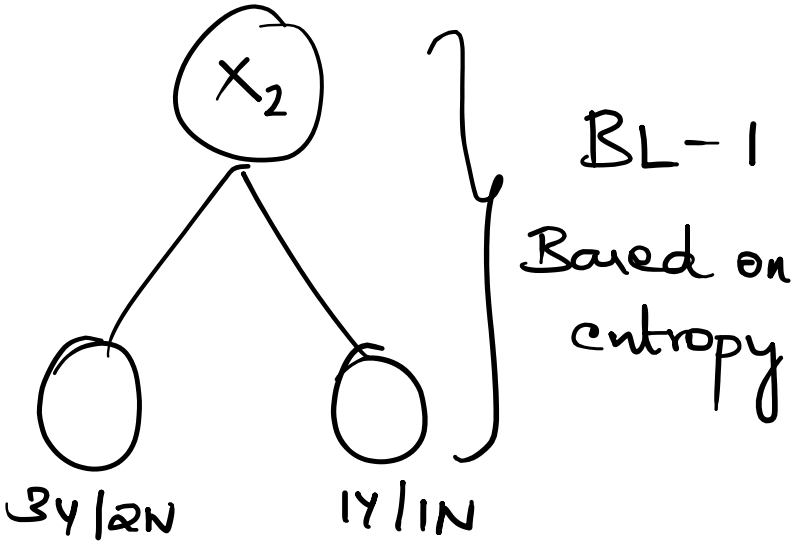
.

.

1/7

Decision stumps





Performance of decision stump (BL-1):

$$\Rightarrow \frac{1}{2} \log_e \left(\frac{(1-\epsilon_t)}{\epsilon_t} \right)$$

$$\Rightarrow \frac{1}{2} \log_e \left(\frac{(1-1/7)}{(1/7)} \right) \Rightarrow 0.89$$

Weight update :

$$\Rightarrow w_i^{(t+1)} = w_i^{(t)} \cdot e^{\alpha_t}$$

$$= 1/7 \times e^{0.89}$$

$$= 0.35$$

$$= w_i^{(t)} \cdot e^{-\alpha t}$$

$$= 1/7 \times e^{-0.89}$$

$$= 0.05$$

$0.35 / 0.65 \Rightarrow$

X1	X2	X3	Y
1			
2			
3			
4			
5			
6			×
7			

Sample Weight	updated Weight	Normalized
1/7	0.05	0.076
.	0.05	0.076
.	0.05	0.076
.	0.05	0.076
.	0.05	0.076
.	0.35	0.538
1/7	0.05	0.076

X1	X2	X3	Y	Weights
1				1/7
2				1/7
3				1/7
4				1/7
5				1/7
6			X	1/7
7				1/7

Normalized Weights	Bucket
0.07	0 to 0.07
0.07	0.07 to 0.14
0.07	0.14 to 0.21
0.07	0.21 to 0.28
0.07	0.28 to 0.35
0.53	0.35 to 0.88
0.07	0.88 to 0.95